

Linux Installation and
ROS 2 Humble Setup

Complete Installation Report

Operating System	Ubuntu 22.04.5 LTS (Jammy Jellyfish)
Virtualization	VMware Workstation
ROS Distribution	ROS 2 Humble
Architecture	64-bit (amd64)
Final Status	Installed and verified successfully

1. Objective

The objective of this task was to install a Linux operating system, configure ROS 2 Humble, verify that ROS is working correctly, and document the complete process. Ubuntu 22.04.5 LTS was selected because ROS 2 Humble is designed for Ubuntu 22.04 (Jammy). VMware Workstation was used to run Ubuntu as a virtual machine without modifying the main Windows installation.

2. Software and Requirements

- VMware Workstation installed on the Windows host computer.
- Ubuntu 22.04.5 Desktop 64-bit ISO image.
- Internet connection for Ubuntu package downloads.
- Approximately 3 GB or more of additional free space for ROS 2 Humble Desktop packages.

3. Downloading Ubuntu

The Ubuntu 22.04.5 Desktop 64-bit ISO file was downloaded first. The downloaded file was then used as the installation image inside VMware Workstation.

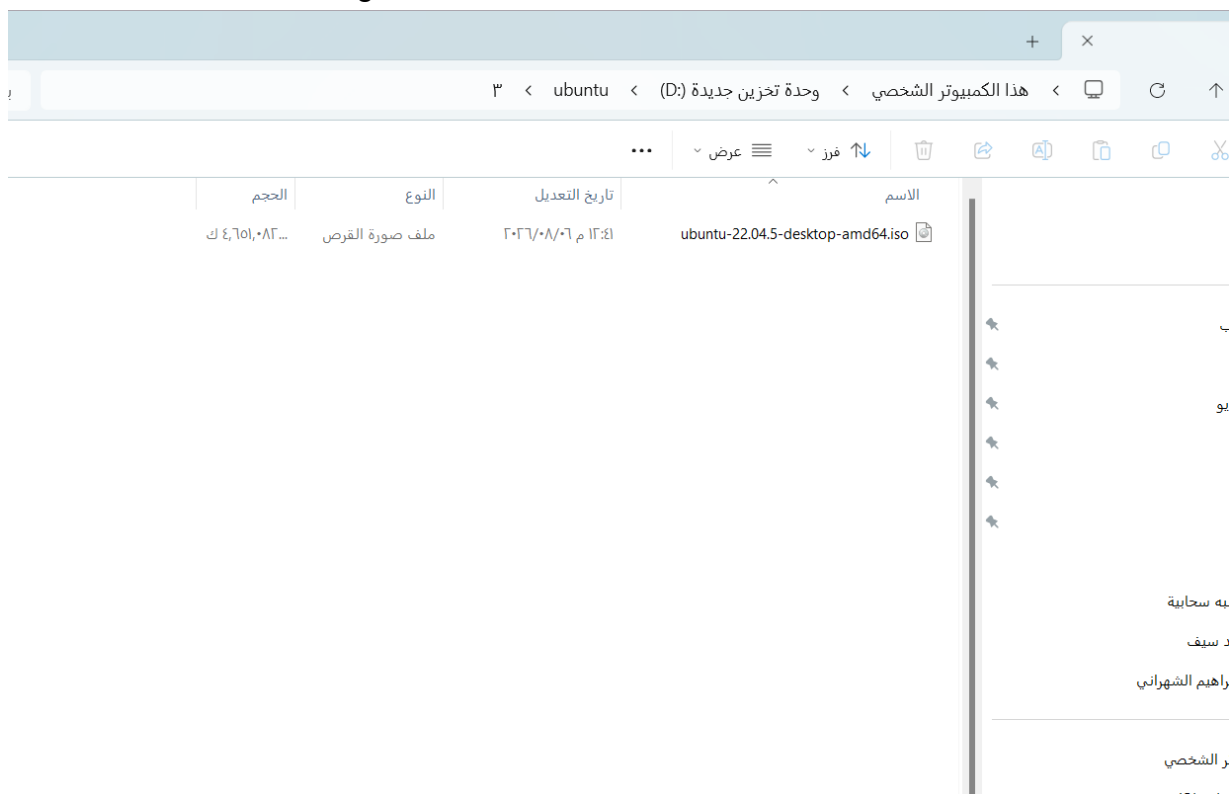


Figure 1. Ubuntu 22.04.5 Desktop ISO downloaded and ready for VMware.

4. Creating the Ubuntu Virtual Machine

A new virtual machine was created in VMware using the Typical (recommended) configuration.

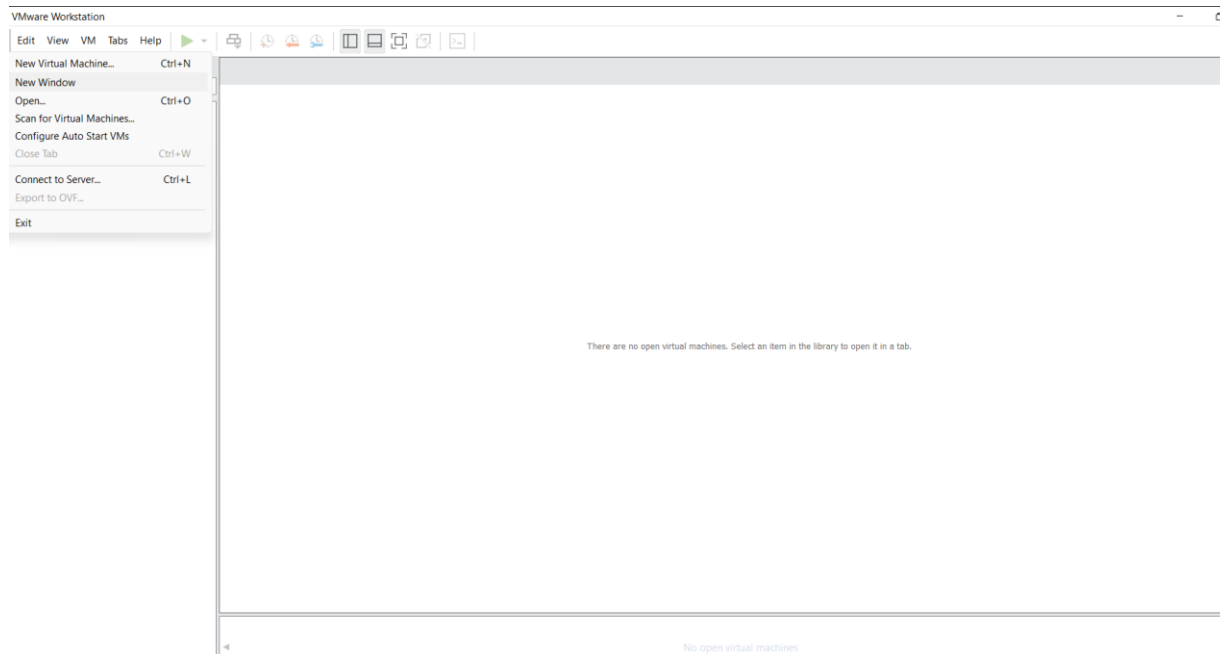


Figure 2. VMware Workstation before creating the new virtual machine.

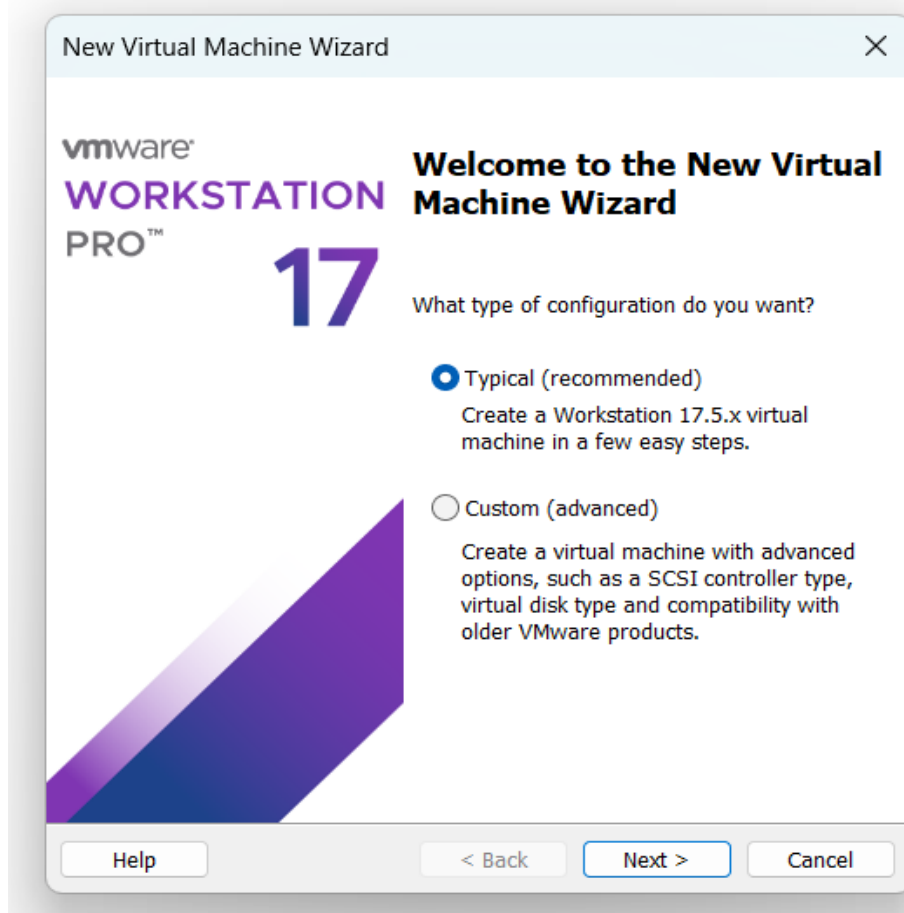


Figure 3. New Virtual Machine Wizard with Typical configuration selected.

4.1 Selecting the Ubuntu ISO

The downloaded Ubuntu ISO was selected as the installer disc image. VMware detected Ubuntu 64-bit 22.04.5 and prepared the installation.

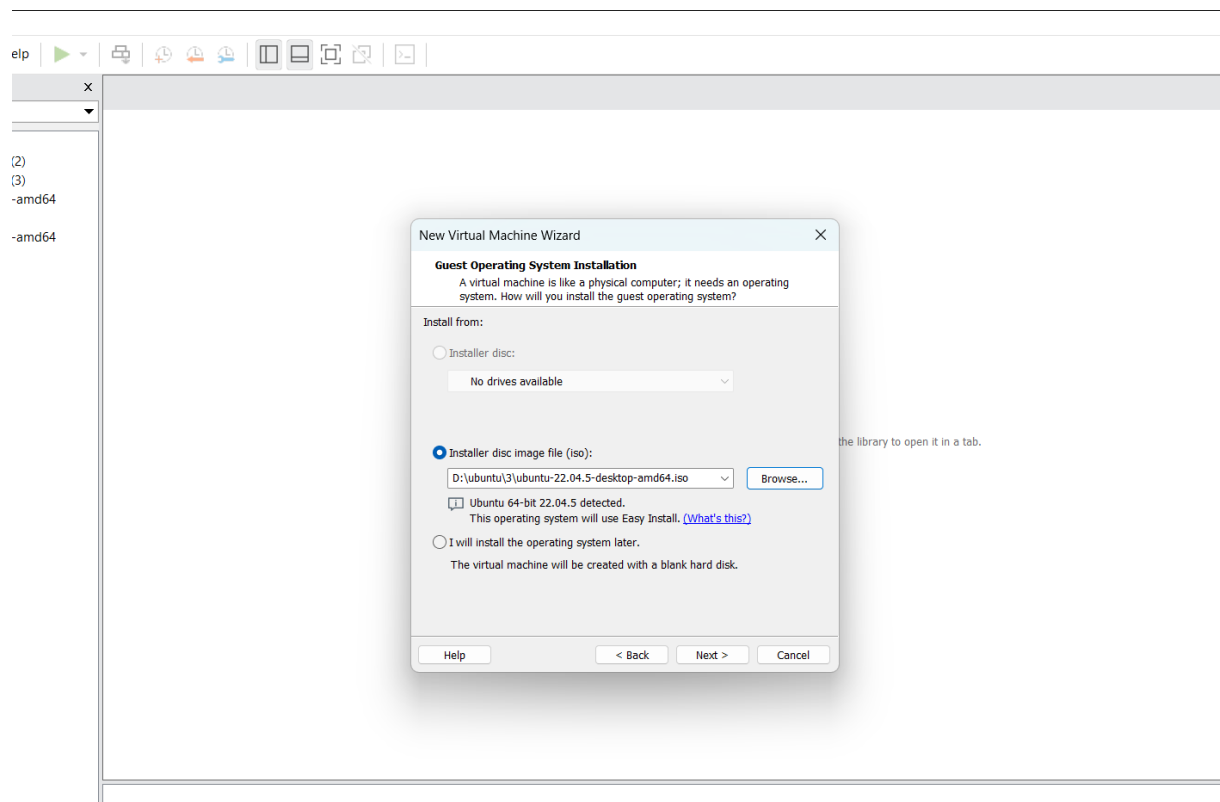


Figure 4. Ubuntu 22.04.5 ISO selected in VMware.

5. Installing Ubuntu 22.04.5 LTS

During the Ubuntu installer, English (US) was kept as the keyboard layout. A normal installation was selected, and the virtual disk was used for Ubuntu.

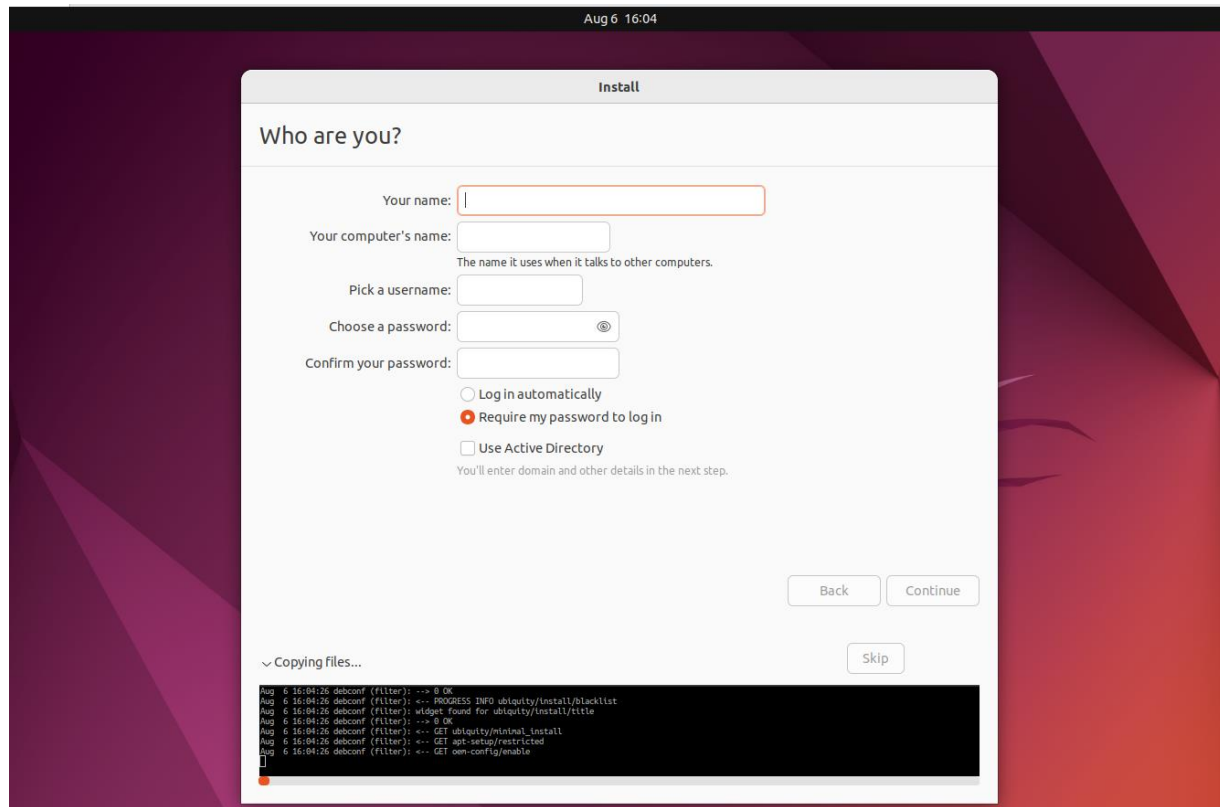


Figure 5. Ubuntu keyboard layout selection.

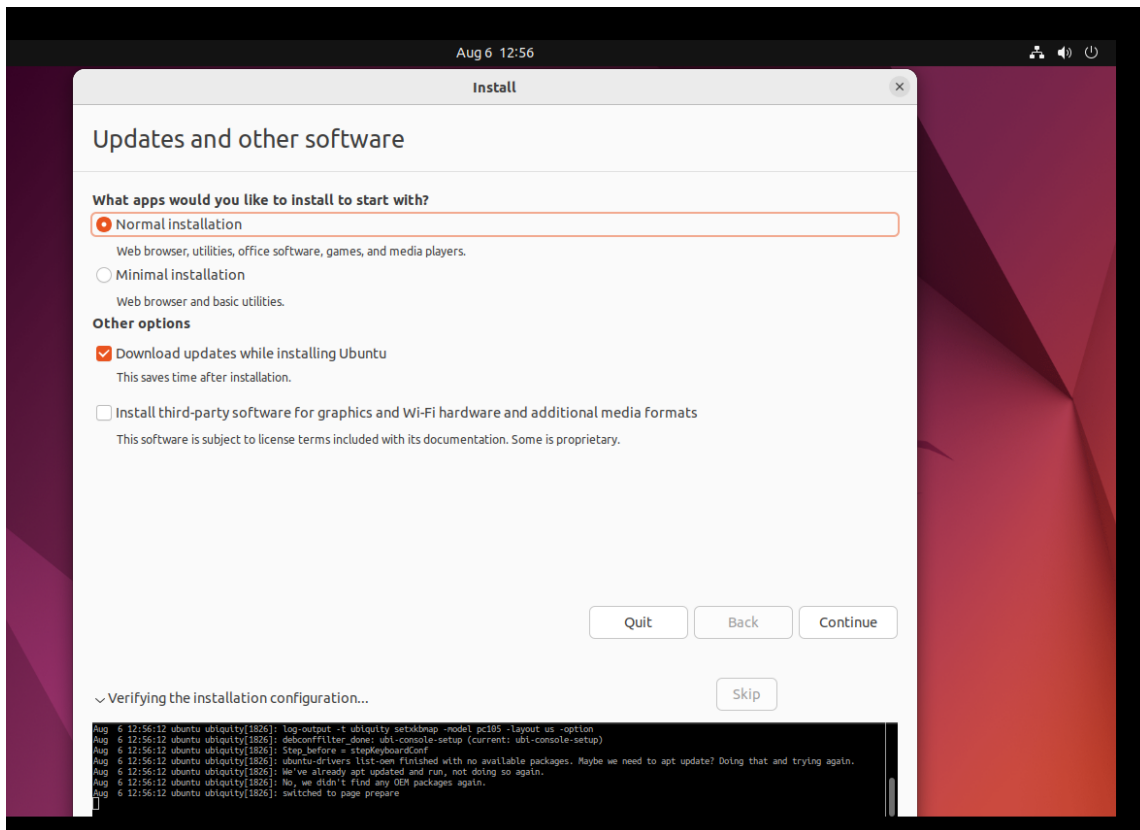


Figure 6. Ubuntu installation type and software options.

The 'Erase disk and install Ubuntu' option applied only to the virtual disk assigned to this VMware virtual machine. It did not erase the Windows host drive.

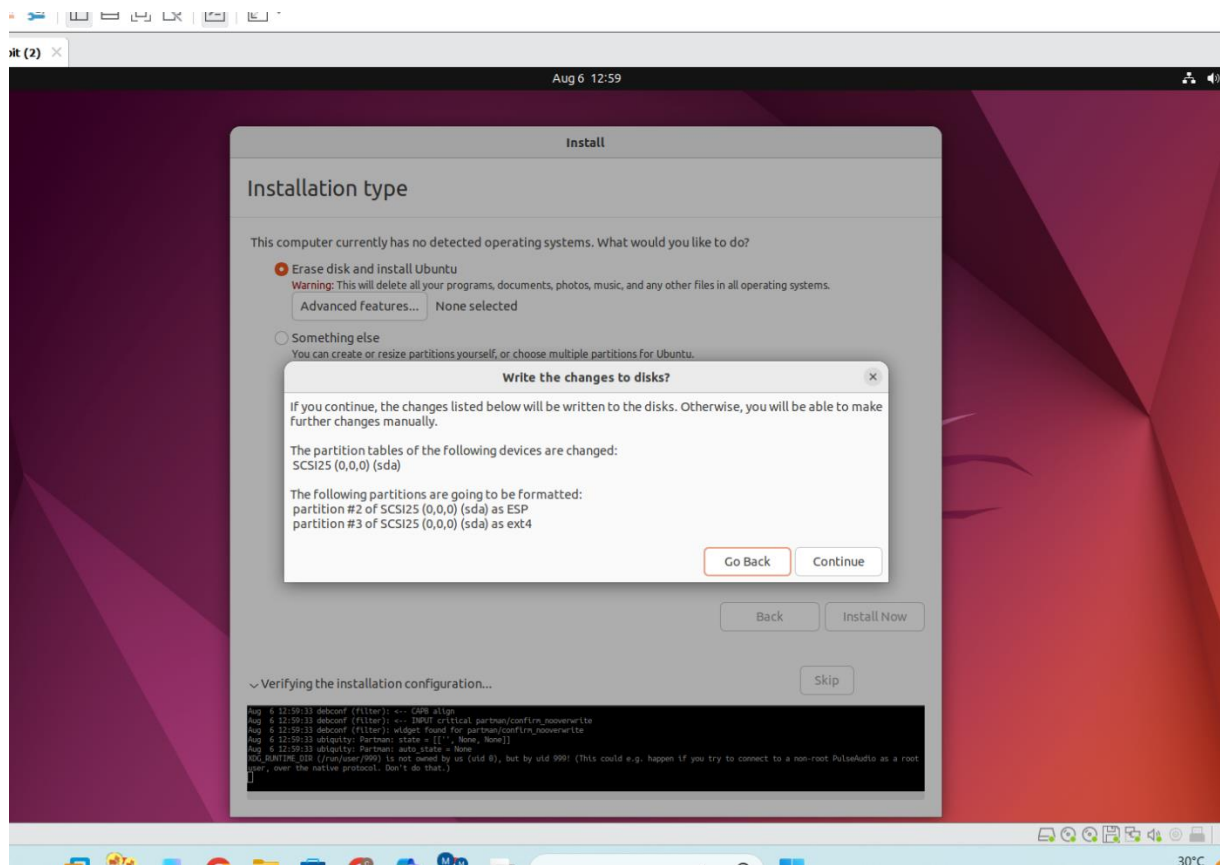


Figure 8. Confirmation to write the partition changes to the virtual disk.

5.1 Creating the Ubuntu User

A local Ubuntu account was created. The username was set to 'ubuntu'. Password-protected login was enabled.

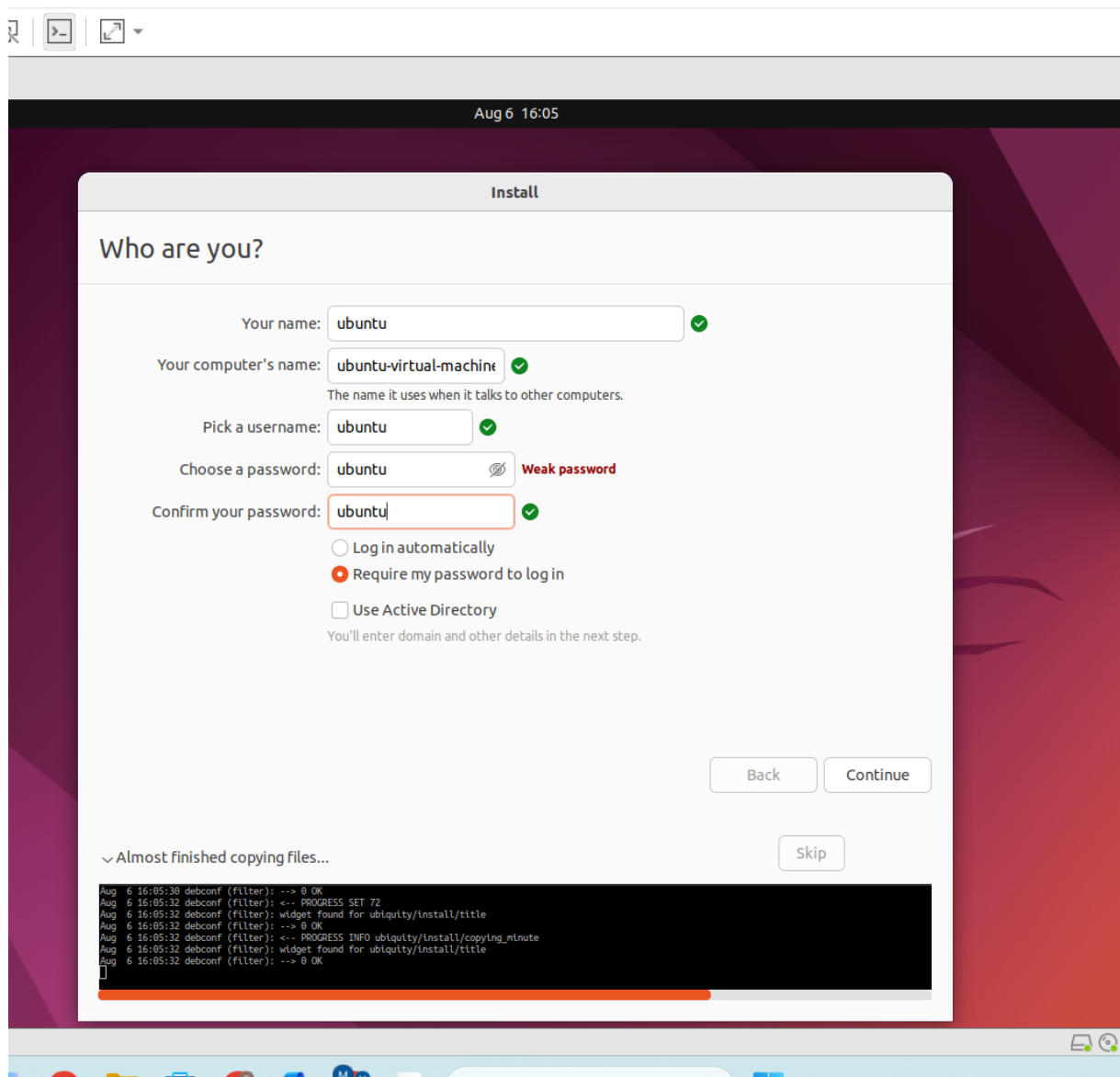


Figure 9. Ubuntu user account configuration.

5.2 Completing the Installation

Ubuntu copied the required files and completed the installation. The virtual machine was then restarted.

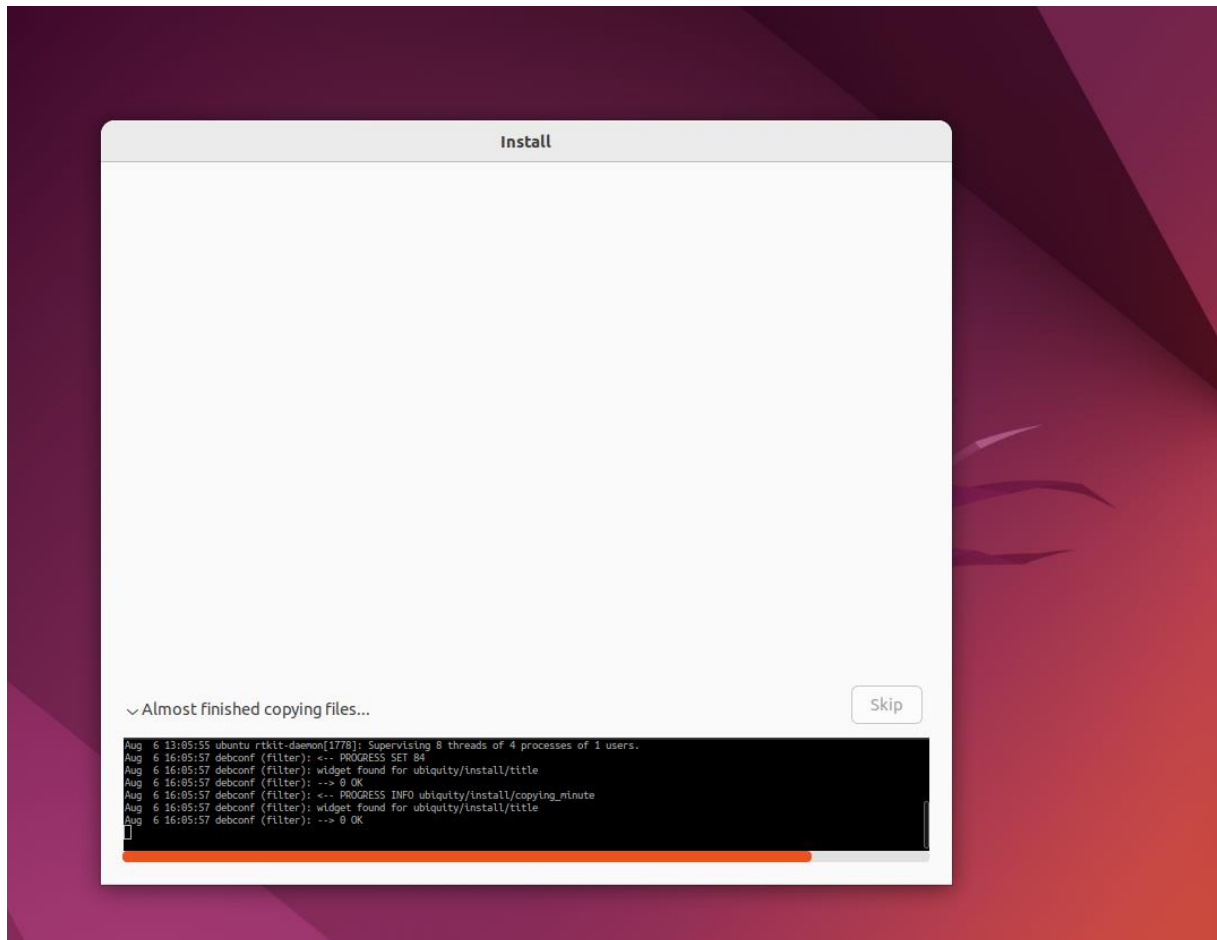


Figure 10. Ubuntu installation in progress.

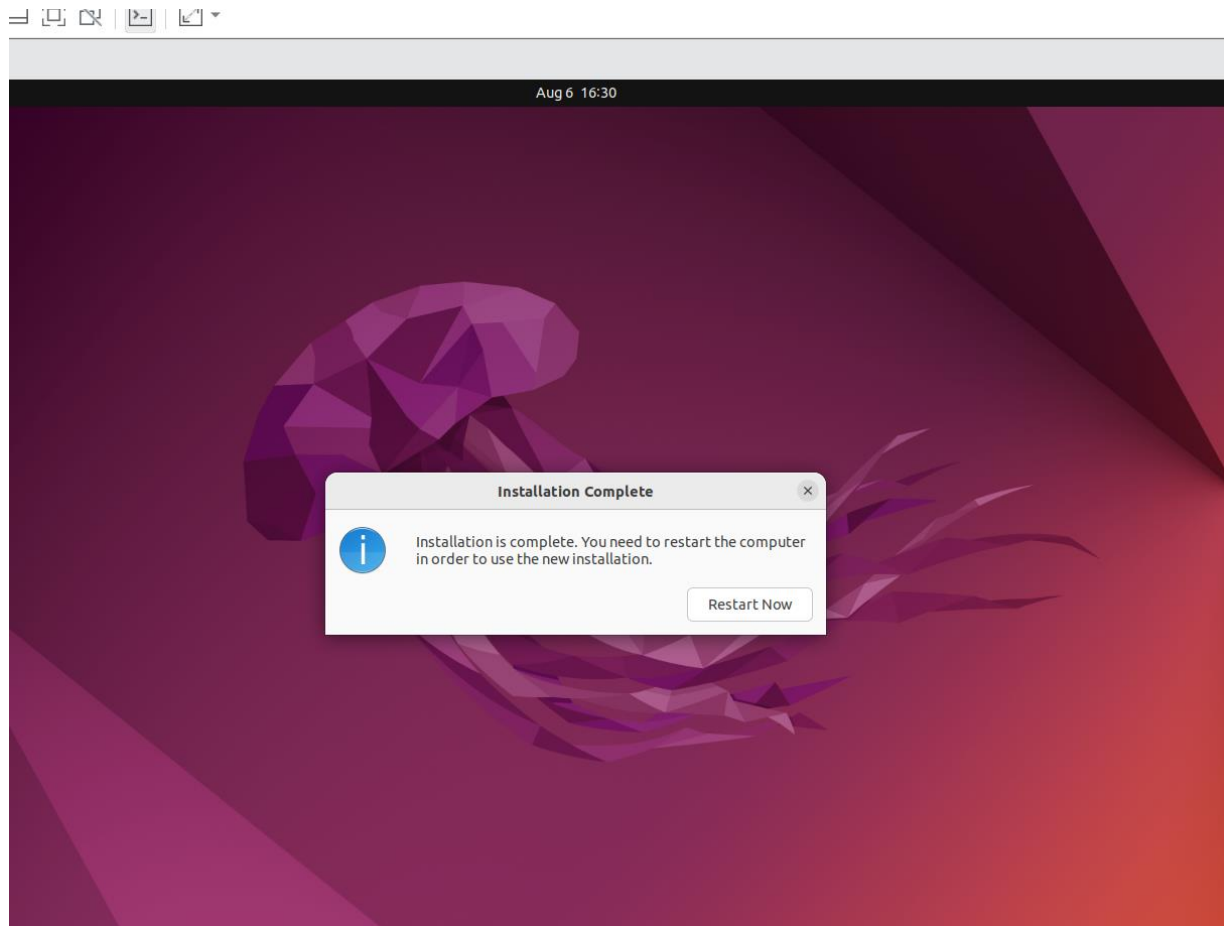


Figure 11. Ubuntu installation completed successfully.

6. Preparing Ubuntu for ROS 2

After Ubuntu started, the terminal was used for all ROS installation commands. The computer hostname was changed to 'ubuntu' so that the terminal prompt became easier to read.

```
sudo hostnamectl set-hostname ubuntu
```

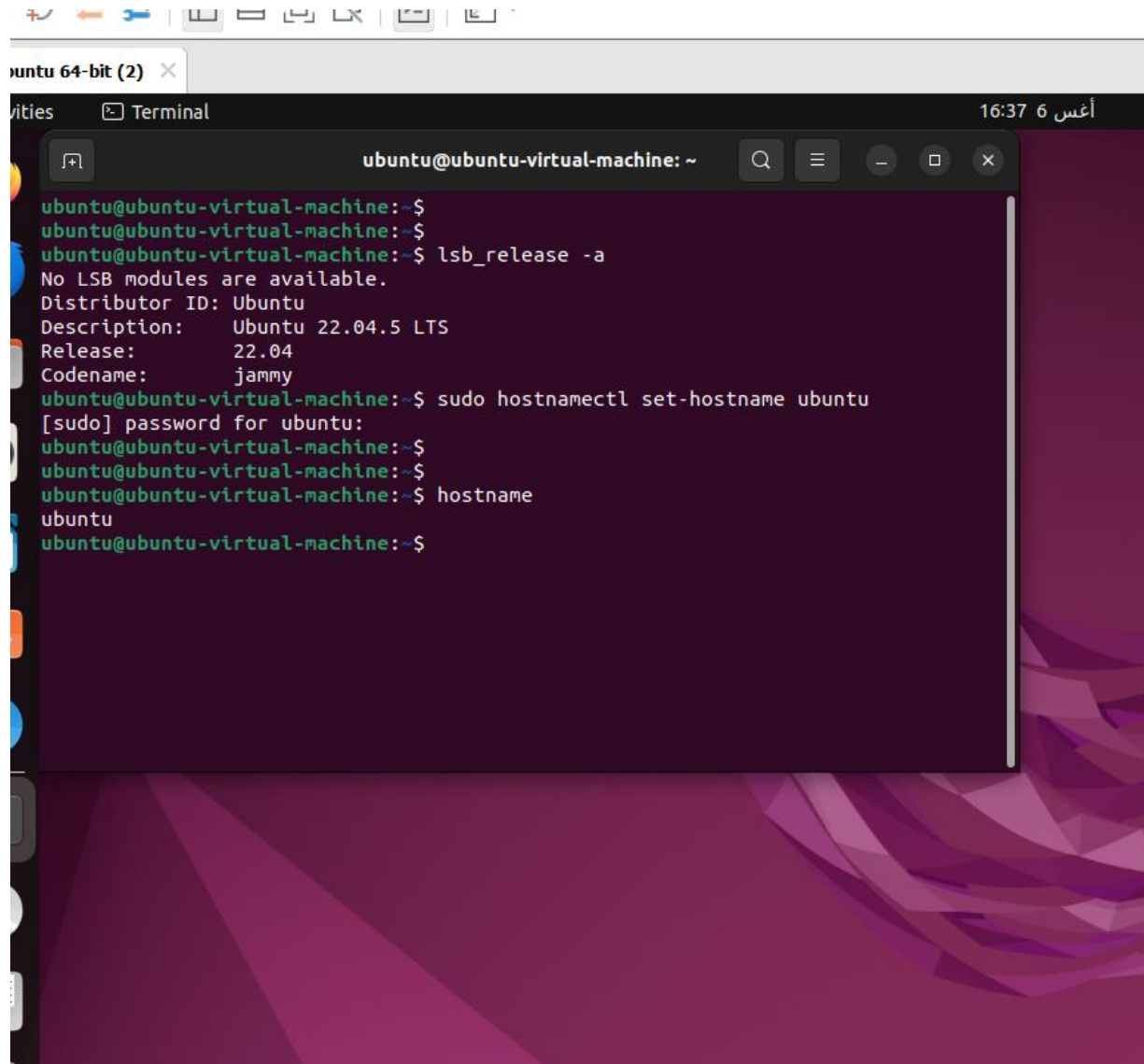


Figure 12. Hostname changed successfully to ubuntu.

6.1 Verifying the Ubuntu Version

```
lsb_release -a
```

The result confirmed Ubuntu 22.04.5 LTS with the Jammy codename.

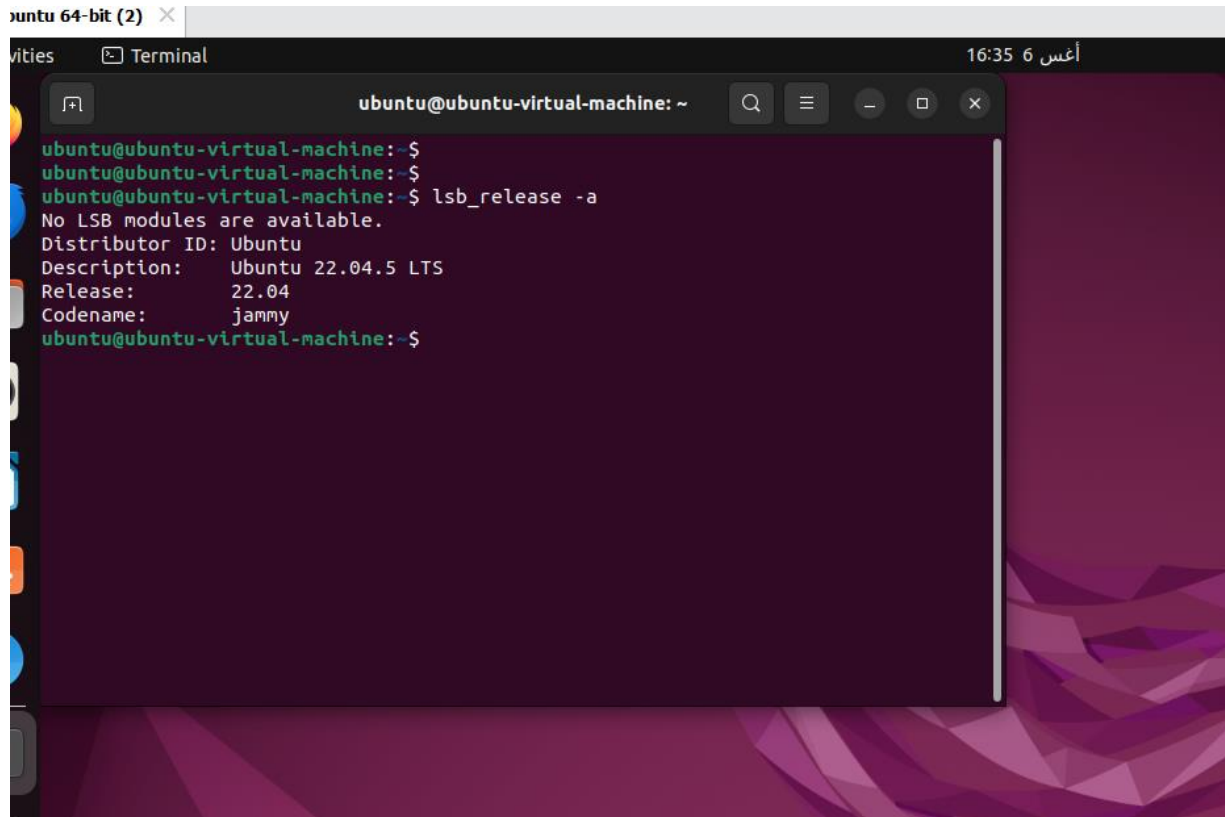
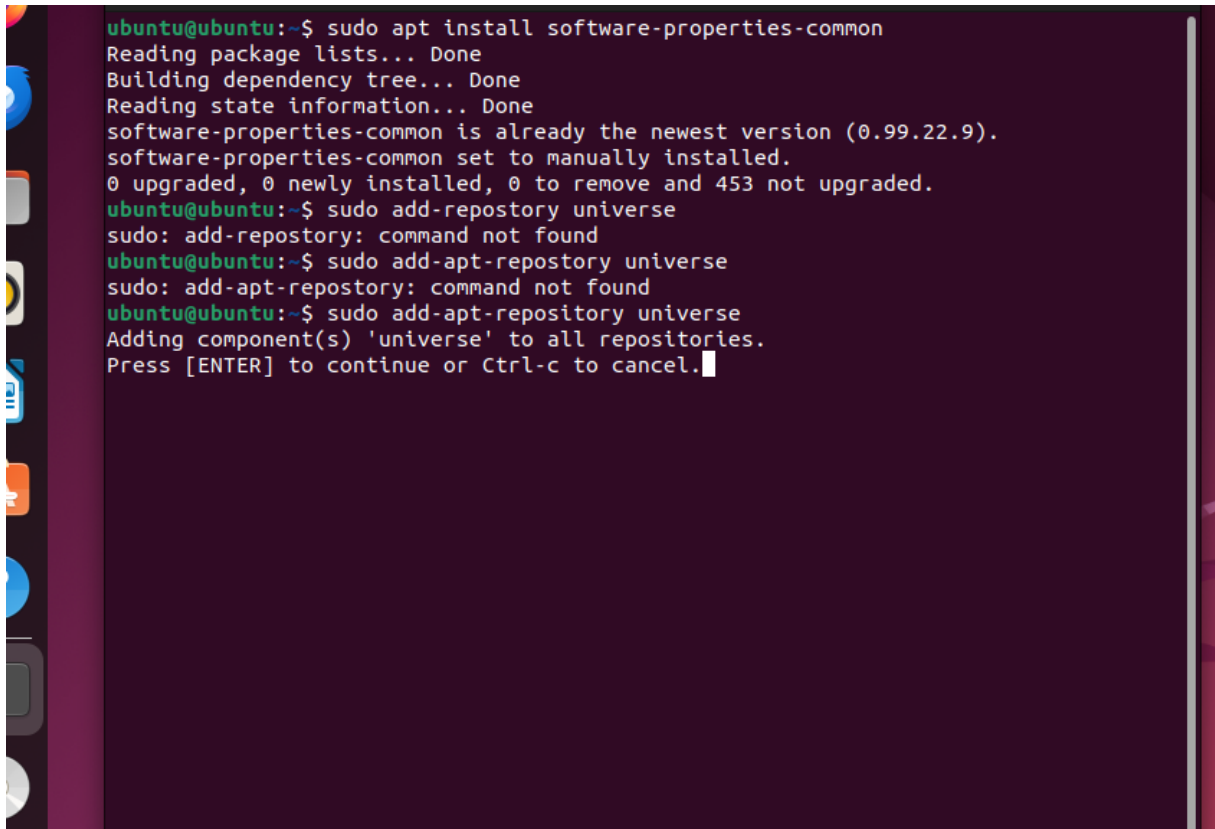
A screenshot of a terminal window titled 'Terminal' with a dark background. The terminal shows the command 'lsb_release -a' being executed. The output displays system information: 'Distributor ID: Ubuntu', 'Description: Ubuntu 22.04.5 LTS', 'Release: 22.04', and 'Codename: jammy'. The prompt 'ubuntu@ubuntu-virtual-machine: ~' is visible at the top of the terminal. The window's title bar includes 'ubuntu 64-bit (2)' and a search icon. The system clock in the top right corner shows '16:35 6 أغسطس'.

Figure 13. Verification of Ubuntu 22.04.5 LTS (Jammy).

6.2 Updating the Package List

```
sudo apt update
```

The package index was updated before installing ROS and its dependencies.

A terminal window with a dark purple background and a Ubuntu desktop sidebar on the left. The terminal shows the output of several commands: 'sudo apt install software-properties-common' which reads package lists, builds a dependency tree, and reads state information, noting that 'software-properties-common' is already the newest version (0.99.22.9) and is set to manually installed. It also reports 0 upgrades, 0 new installations, 0 removals, and 453 not upgraded. Then, 'sudo add-repository universe' is run, resulting in a 'command not found' error. Next, 'sudo add-apt-repository universe' is run, also resulting in a 'command not found' error. Finally, 'sudo add-apt-repository universe' is run again, which successfully adds the 'universe' component to all repositories, with a prompt to press [ENTER] to continue or Ctrl-c to cancel.

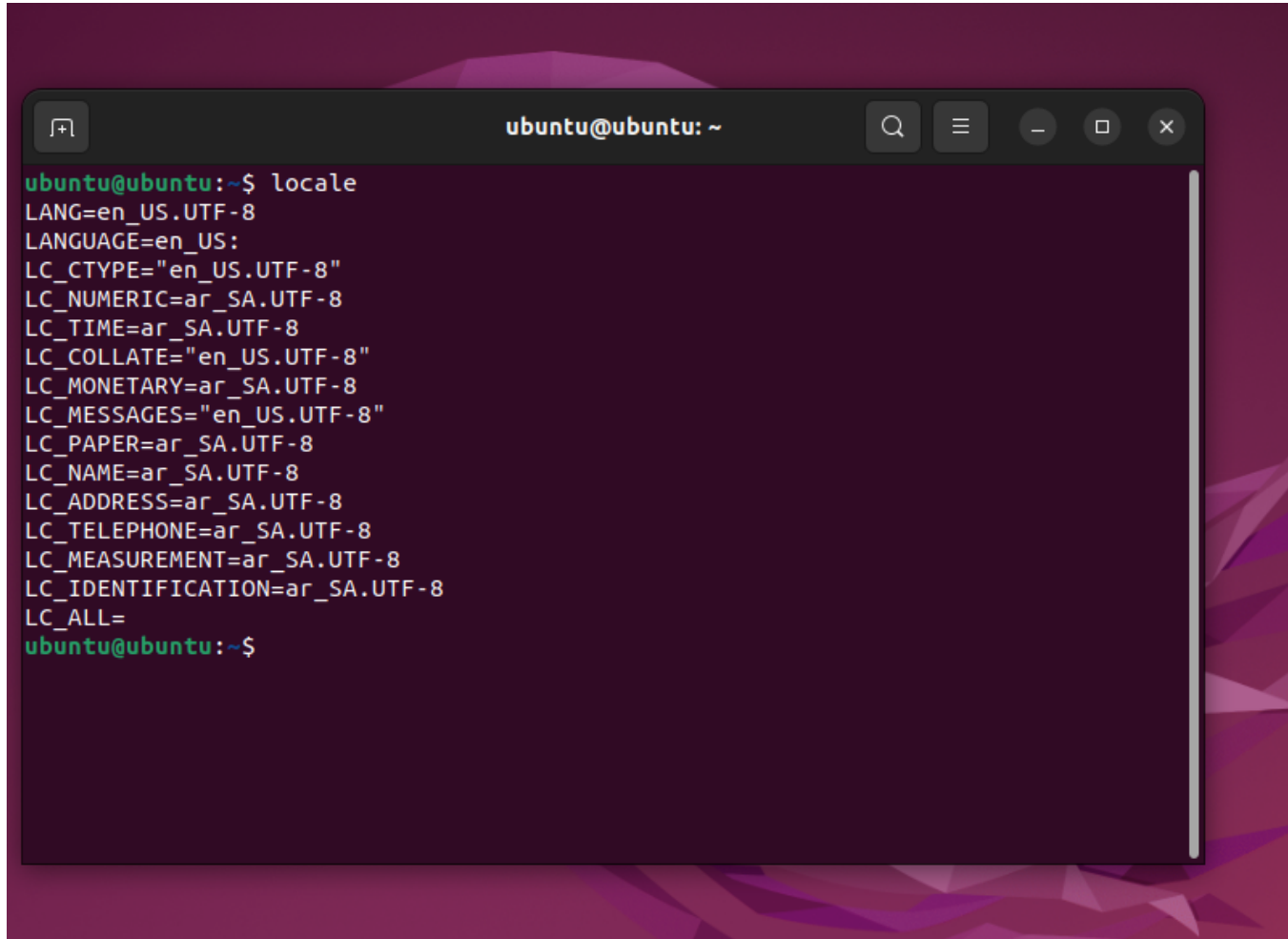
```
ubuntu@ubuntu:~$ sudo apt install software-properties-common
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
software-properties-common is already the newest version (0.99.22.9).
software-properties-common set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 453 not upgraded.
ubuntu@ubuntu:~$ sudo add-repository universe
sudo: add-repository: command not found
ubuntu@ubuntu:~$ sudo add-apt-repository universe
sudo: add-apt-repository: command not found
ubuntu@ubuntu:~$ sudo add-apt-repository universe
Adding component(s) 'universe' to all repositories.
Press [ENTER] to continue or Ctrl-c to cancel.
```

Figure 14. Ubuntu package list updated successfully.

6.3 Checking the Locale

```
locale
```

The system had UTF-8 locale support, including LANG=en_US.UTF-8, which is suitable for ROS 2.

A terminal window titled 'ubuntu@ubuntu: ~' with standard window controls. The terminal displays the output of the 'locale' command. The output shows various locale settings, all of which are configured to use UTF-8 encoding, indicating that the system is ready for ROS 2 installation.

```
ubuntu@ubuntu:~$ locale
LANG=en_US.UTF-8
LANGUAGE=en_US:
LC_CTYPE="en_US.UTF-8"
LC_NUMERIC=ar_SA.UTF-8
LC_TIME=ar_SA.UTF-8
LC_COLLATE="en_US.UTF-8"
LC_MONETARY=ar_SA.UTF-8
LC_MESSAGES="en_US.UTF-8"
LC_PAPER=ar_SA.UTF-8
LC_NAME=ar_SA.UTF-8
LC_ADDRESS=ar_SA.UTF-8
LC_TELEPHONE=ar_SA.UTF-8
LC_MEASUREMENT=ar_SA.UTF-8
LC_IDENTIFICATION=ar_SA.UTF-8
LC_ALL=
ubuntu@ubuntu:~$
```

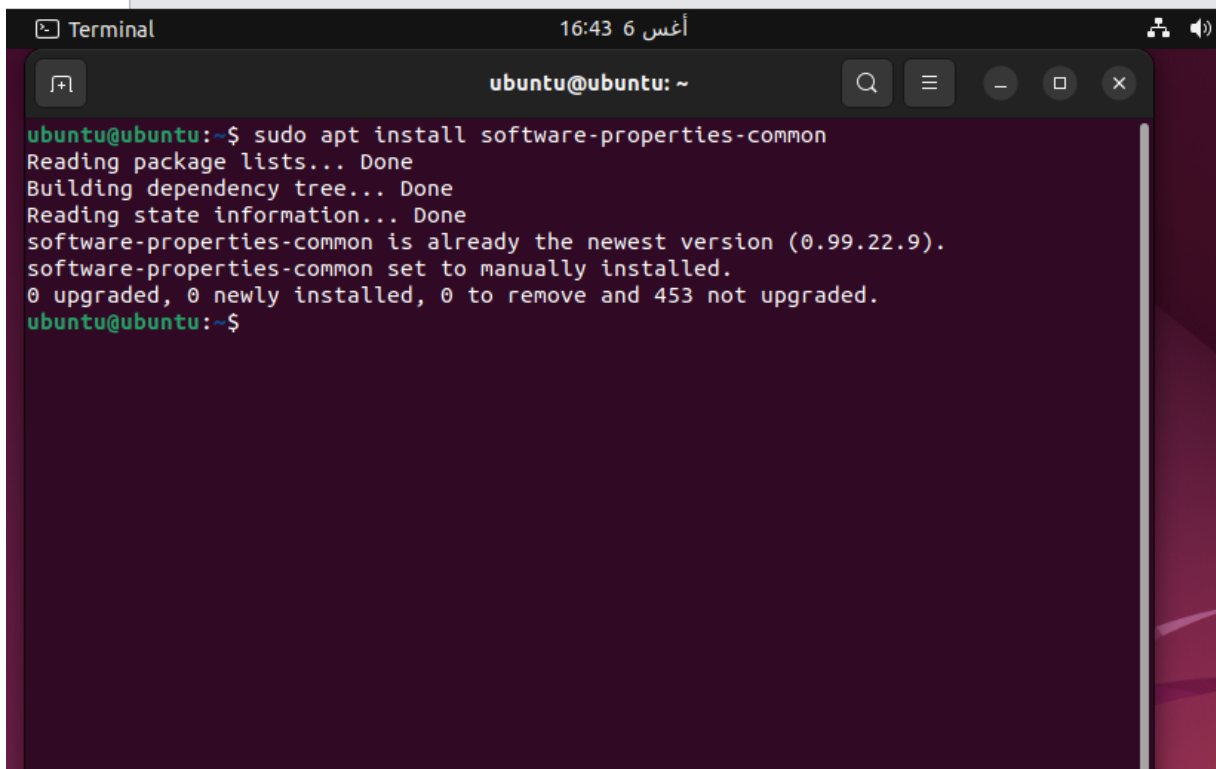
Figure 15. Locale configuration checked before ROS installation.

7. Preparing ROS 2 Repositories

7.1 Installing Repository Tools

```
sudo apt install software-properties-common
```

The package was already installed and up to date.

A terminal window titled 'Terminal' with a dark background and light text. The window shows the command 'sudo apt install software-properties-common' being executed. The output indicates that the package is already installed and up to date. The terminal window has a title bar with the text 'Terminal' and a clock showing '16:43 6 أغسطس'. The prompt is 'ubuntu@ubuntu: ~'.

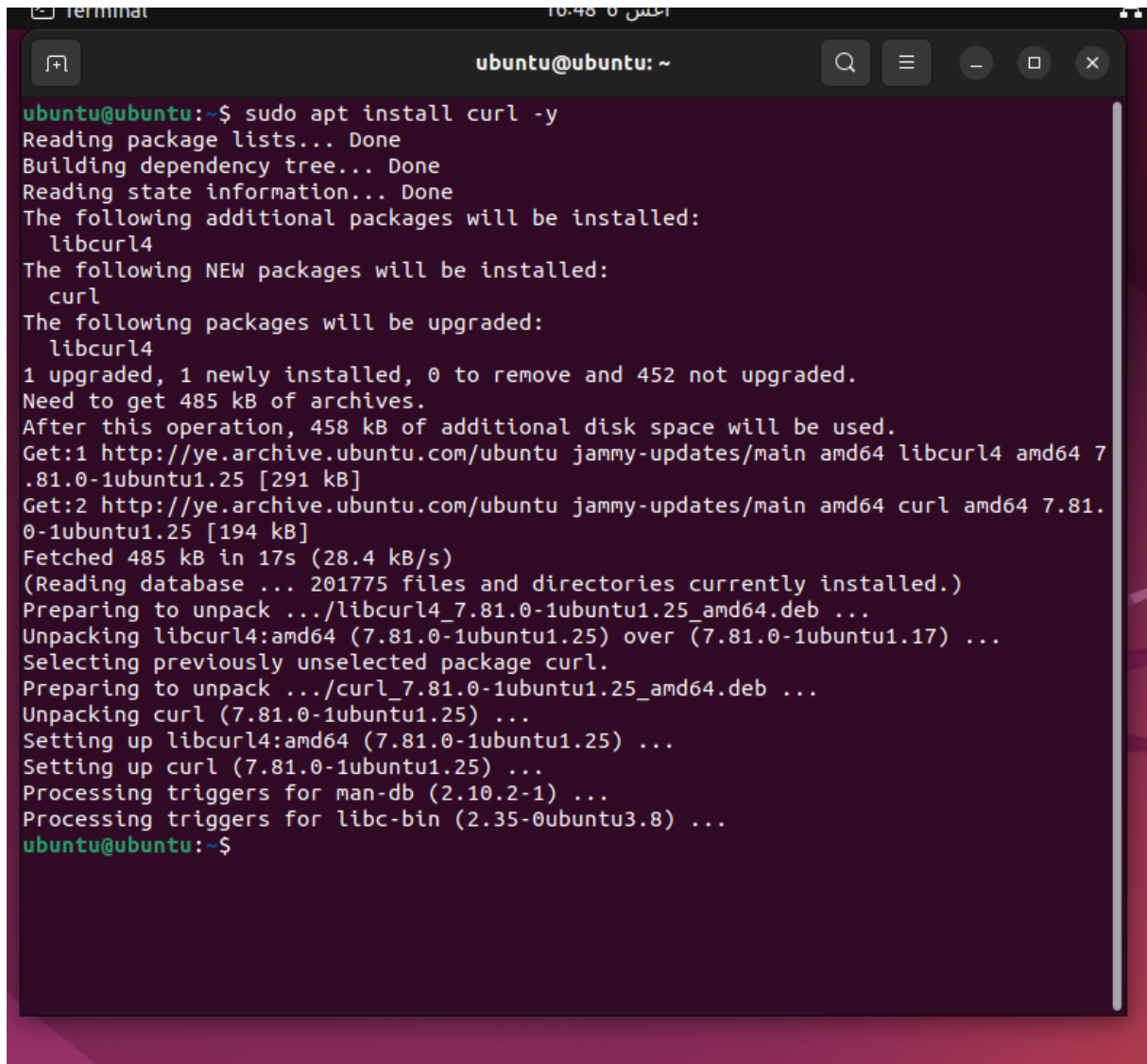
```
ubuntu@ubuntu:~$ sudo apt install software-properties-common
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
software-properties-common is already the newest version (0.99.22.9).
software-properties-common set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 453 not upgraded.
ubuntu@ubuntu:~$
```

Figure 16. software-properties-common checked successfully.

7.2 Enabling the Universe Repository

```
sudo add-apt-repository universe
```

The Universe repository was enabled and the package lists were refreshed.



```
ubuntu@ubuntu: ~$ sudo apt install curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcurl4
The following NEW packages will be installed:
  curl
The following packages will be upgraded:
  libcurl4
1 upgraded, 1 newly installed, 0 to remove and 452 not upgraded.
Need to get 485 kB of archives.
After this operation, 458 kB of additional disk space will be used.
Get:1 http://ye.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libcurl4 amd64 7.81.0-1ubuntu1.25 [291 kB]
Get:2 http://ye.archive.ubuntu.com/ubuntu jammy-updates/main amd64 curl amd64 7.81.0-1ubuntu1.25 [194 kB]
Fetched 485 kB in 17s (28.4 kB/s)
(Reading database ... 201775 files and directories currently installed.)
Preparing to unpack .../libcurl4_7.81.0-1ubuntu1.25_amd64.deb ...
Unpacking libcurl4:amd64 (7.81.0-1ubuntu1.25) over (7.81.0-1ubuntu1.17) ...
Selecting previously unselected package curl.
Preparing to unpack .../curl_7.81.0-1ubuntu1.25_amd64.deb ...
Unpacking curl (7.81.0-1ubuntu1.25) ...
Setting up libcurl4:amd64 (7.81.0-1ubuntu1.25) ...
Setting up curl (7.81.0-1ubuntu1.25) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
ubuntu@ubuntu:~$
```

Figure 17. Universe repository enabled.

7.3 Installing curl

```
sudo apt install curl -y
```

curl was installed to download the ROS repository signing key.

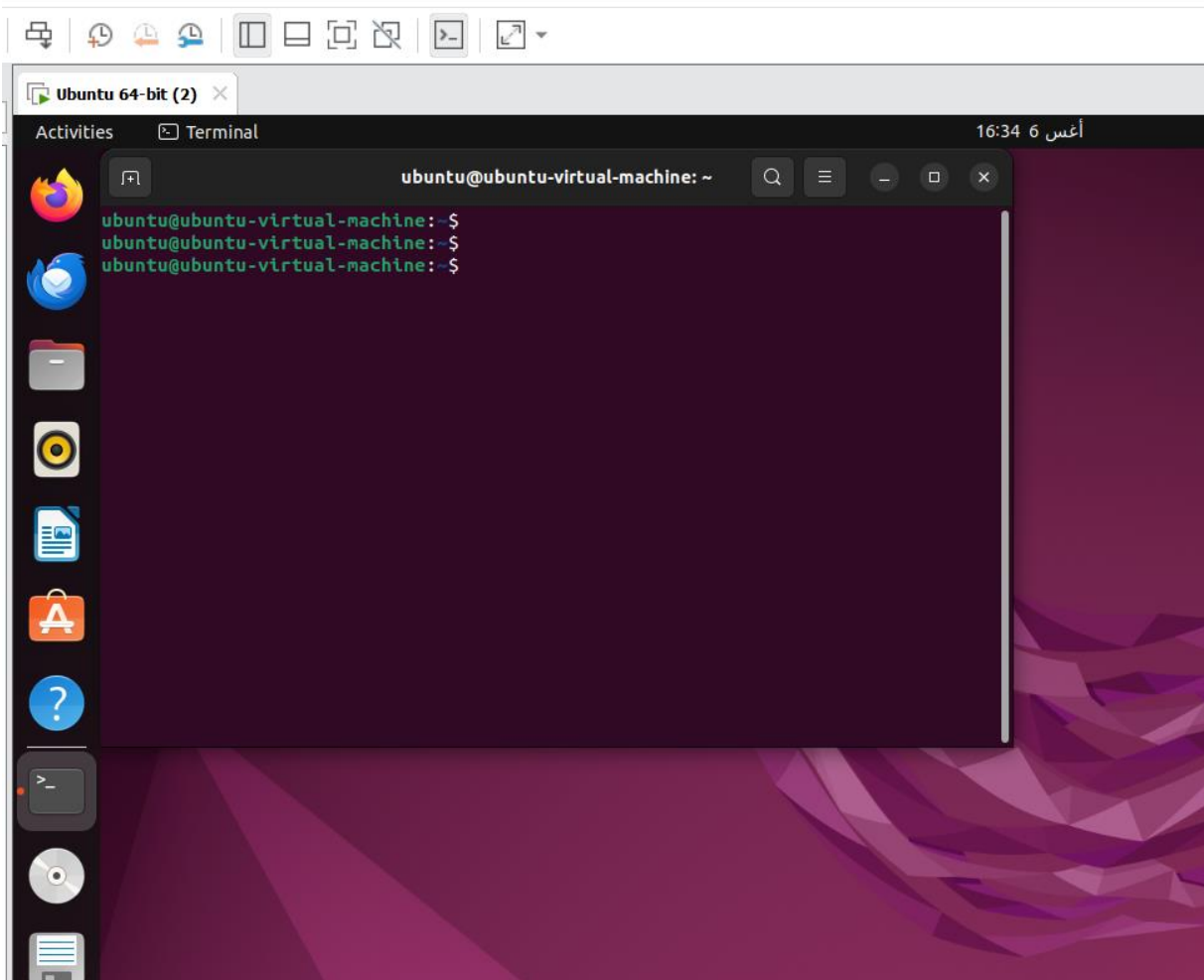
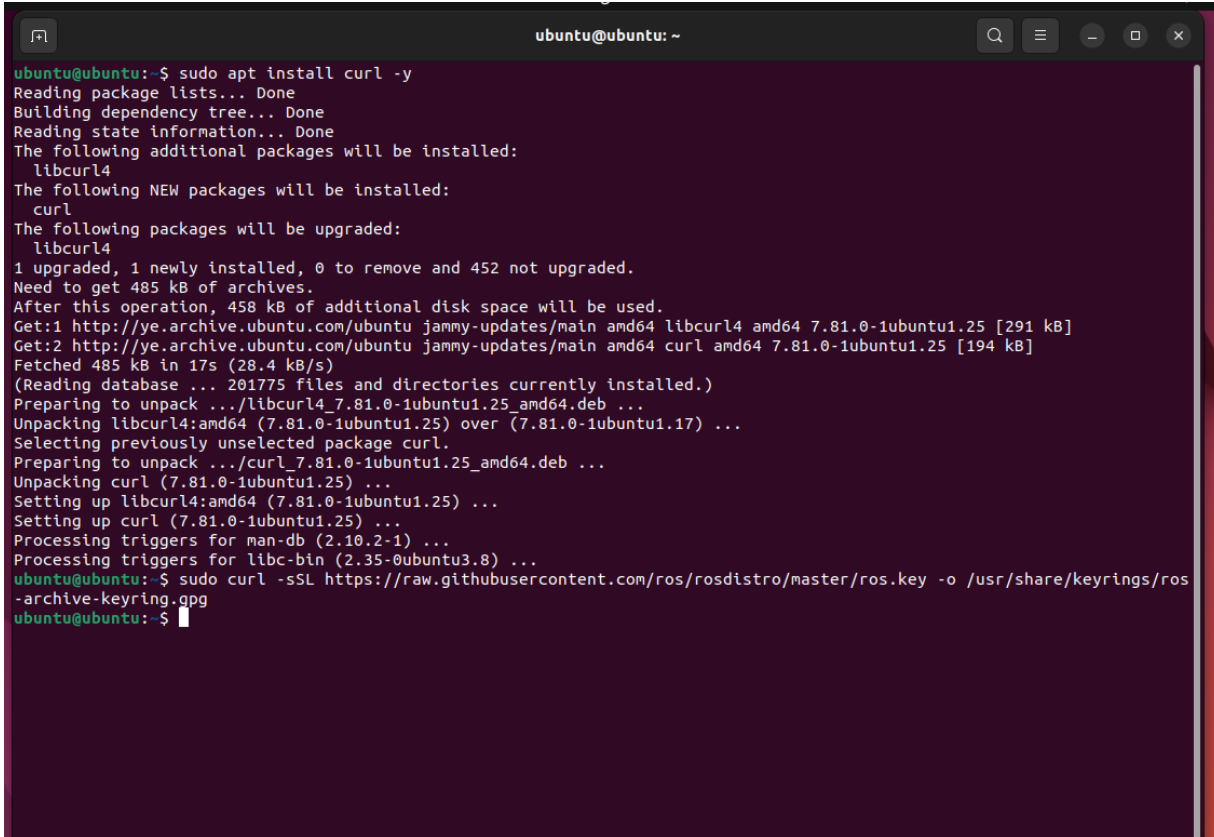


Figure 18. curl installed successfully.

7.4 Adding the ROS Repository Key

```
sudo curl -sSL  
https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o  
/usr/share/keyrings/ros-archive-keyring.gpg
```

The ROS signing key was saved in the Ubuntu keyrings directory.

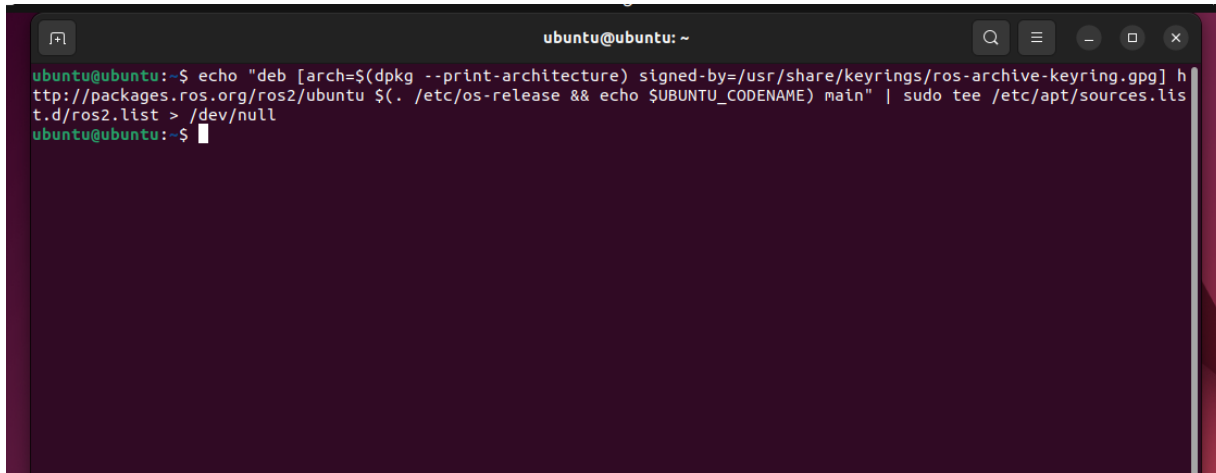
A terminal window titled 'ubuntu@ubuntu: ~' showing the installation of curl and the execution of a command to add the ROS repository key. The terminal output shows the following steps: 1. Running 'sudo apt install curl -y'. 2. Reading package lists, building dependency tree, and reading state information. 3. Listing additional packages to be installed (libcurl4) and NEW packages (curl). 4. Listing packages to be upgraded (libcurl4). 5. Showing disk space requirements and sources. 6. Fetching the curl package. 7. Unpacking and setting up libcurl4 and curl. 8. Processing triggers for man-db and libc-bin. 9. Running the command 'sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o /usr/share/keyrings/ros-archive-keyring.gpg'.

```
ubuntu@ubuntu:~$ sudo apt install curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcurl4
The following NEW packages will be installed:
  curl
The following packages will be upgraded:
  libcurl4
1 upgraded, 1 newly installed, 0 to remove and 452 not upgraded.
Need to get 485 kB of archives.
After this operation, 458 kB of additional disk space will be used.
Get:1 http://ye.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libcurl4 amd64 7.81.0-1ubuntu1.25 [291 kB]
Get:2 http://ye.archive.ubuntu.com/ubuntu jammy-updates/main amd64 curl amd64 7.81.0-1ubuntu1.25 [194 kB]
Fetched 485 kB in 17s (28.4 kB/s)
(Reading database ... 201775 files and directories currently installed.)
Preparing to unpack .../libcurl4_7.81.0-1ubuntu1.25_amd64.deb ...
Unpacking libcurl4:amd64 (7.81.0-1ubuntu1.25) over (7.81.0-1ubuntu1.17) ...
Selecting previously unselected package curl.
Preparing to unpack .../curl_7.81.0-1ubuntu1.25_amd64.deb ...
Unpacking curl (7.81.0-1ubuntu1.25) ...
Setting up libcurl4:amd64 (7.81.0-1ubuntu1.25) ...
Setting up curl (7.81.0-1ubuntu1.25) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
ubuntu@ubuntu:~$ sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o /usr/share/keyrings/ros-
archive-keyring.gpg
ubuntu@ubuntu:~$
```

Figure 19. ROS repository signing key added.

7.5 Adding the ROS 2 Package Repository

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo $UBUNTU_CODENAME) main" | sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
```

A terminal window with a dark purple background and light green text. The window title is 'ubuntu@ubuntu: ~'. The command being executed is: `ubuntu@ubuntu:~$ echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo $UBUNTU_CODENAME) main" | sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null`. The output of the command is shown on the next line: `ubuntu@ubuntu:~$`.

```
ubuntu@ubuntu:~$ echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo $UBUNTU_CODENAME) main" | sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
ubuntu@ubuntu:~$
```

Figure 20. ROS 2 repository configuration command executed.

7.6 Updating After Adding ROS

```
sudo apt update
```

The output showed `packages.ros.org/ros2/ubuntu` for Jammy, confirming that the ROS repository was recognized correctly.

A terminal window titled 'Terminal' with a dark background and light text. The window shows the execution of a command to add a ROS 2 repository and the subsequent 'sudo apt update' command. The output of 'apt update' shows several 'Hit' messages for various Ubuntu repositories and a 'Get' message for the ROS 2 repository, indicating it was successfully added and updated. The terminal also shows the package lists being read and the dependency tree being built.

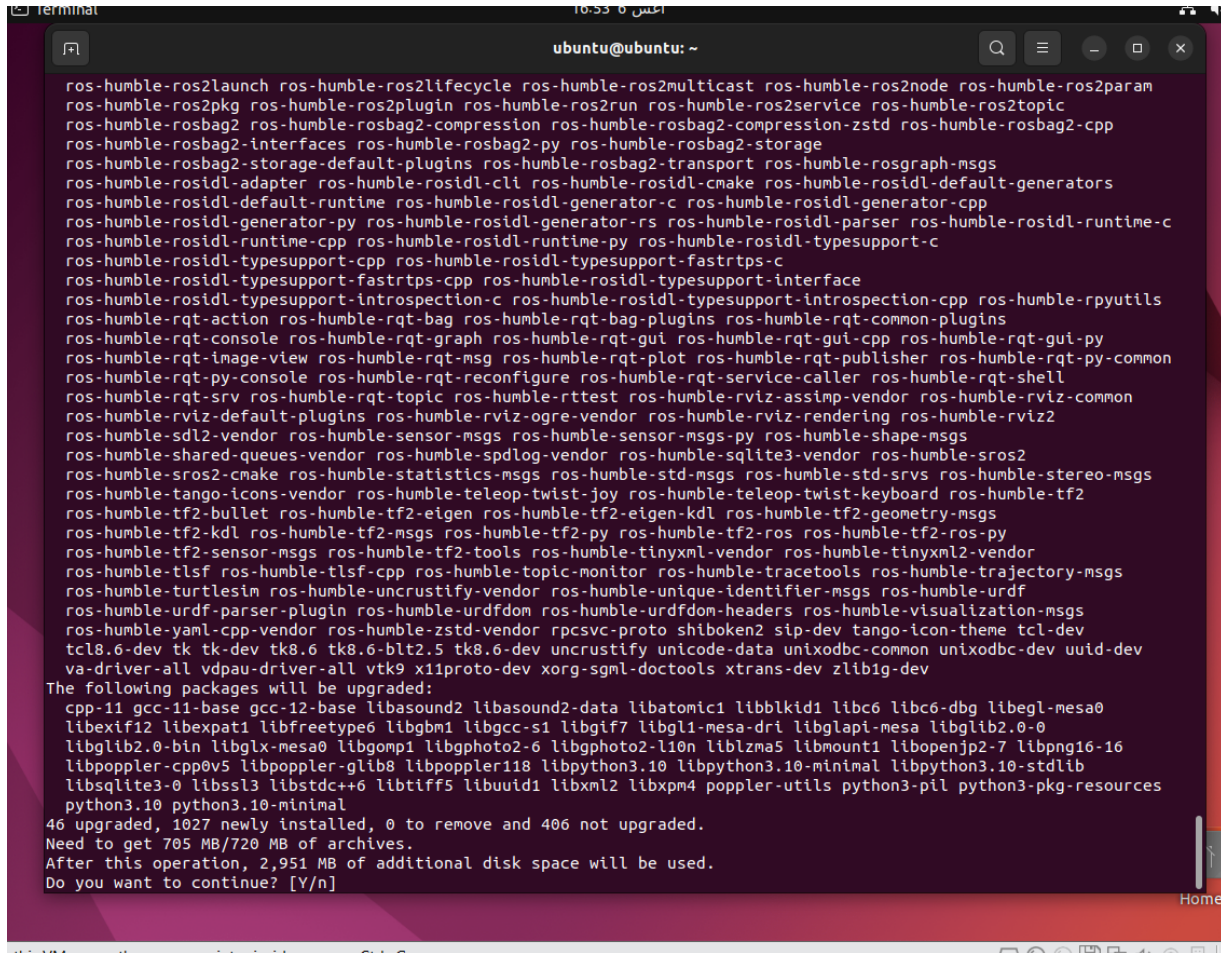
```
Terminal 16:52 6 أغسطس
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo $UBUNTU_CODENAME) main" | sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
ubuntu@ubuntu:~$ sudo apt update
Hit:1 http://ye.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://ye.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://ye.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Get:5 http://packages.ros.org/ros2/ubuntu jammy InRelease [4,682 B]
Get:6 http://packages.ros.org/ros2/ubuntu jammy/main amd64 Packages [1,822 kB]
Fetched 1,827 kB in 42s (43.4 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
452 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu:~$
```

Figure 21. ROS 2 package repository detected successfully.

8. Installing ROS 2 Humble Desktop

```
sudo apt install ros-humble-desktop
```

The installer reported that more than one thousand new packages were required and approximately 2.95 GB of additional disk space would be used. The installation was confirmed by entering Y.



```
ros-humble-ros2launch ros-humble-ros2lifecycle ros-humble-ros2multicast ros-humble-ros2node ros-humble-ros2param
ros-humble-ros2pkg ros-humble-ros2plugin ros-humble-ros2run ros-humble-ros2service ros-humble-ros2topic
ros-humble-rosbag2 ros-humble-rosbag2-compression ros-humble-rosbag2-compression-zstd ros-humble-rosbag2-cpp
ros-humble-rosbag2-interfaces ros-humble-rosbag2-py ros-humble-rosbag2-storage
ros-humble-rosbag2-storage-default-plugins ros-humble-rosbag2-transport ros-humble-rosgraph-msgs
ros-humble-rosidl-adapter ros-humble-rosidl-cli ros-humble-rosidl-cmake ros-humble-rosidl-default-generators
ros-humble-rosidl-default-runtime ros-humble-rosidl-generator-c ros-humble-rosidl-generator-cpp
ros-humble-rosidl-generator-py ros-humble-rosidl-generator-rs ros-humble-rosidl-parser ros-humble-rosidl-runtime-c
ros-humble-rosidl-runtime-cpp ros-humble-rosidl-runtime-py ros-humble-rosidl-typesupport-c
ros-humble-rosidl-typesupport-cpp ros-humble-rosidl-typesupport-fastrtps-c
ros-humble-rosidl-typesupport-fastrtps-cpp ros-humble-rosidl-typesupport-interface
ros-humble-rosidl-typesupport-introspection-c ros-humble-rosidl-typesupport-introspection-cpp ros-humble-rpyutils
ros-humble-rqt-action ros-humble-rqt-bag ros-humble-rqt-bag-plugins ros-humble-rqt-common-plugins
ros-humble-rqt-console ros-humble-rqt-graph ros-humble-rqt-gui ros-humble-rqt-gui-cpp ros-humble-rqt-gui-py
ros-humble-rqt-image-view ros-humble-rqt-msg ros-humble-rqt-plot ros-humble-rqt-publisher ros-humble-rqt-py-common
ros-humble-rqt-py-console ros-humble-rqt-reconfigure ros-humble-rqt-service-caller ros-humble-rqt-shell
ros-humble-rqt-srv ros-humble-rqt-topic ros-humble-rttest ros-humble-rviz-assimp-vendor ros-humble-rviz-common
ros-humble-rviz-default-plugins ros-humble-rviz-ogre-vendor ros-humble-rviz-rendering ros-humble-rviz2
ros-humble-sdl2-vendor ros-humble-sensor-msgs ros-humble-sensor-msgs-py ros-humble-shape-msgs
ros-humble-shared-queues-vendor ros-humble-spdlog-vendor ros-humble-sqlite3-vendor ros-humble-sros2
ros-humble-sros2-cmake ros-humble-statistics-msgs ros-humble-std-msgs ros-humble-std-srvs ros-humble-stereo-msgs
ros-humble-tango-icons-vendor ros-humble-teleop-twist-joy ros-humble-teleop-twist-keyboard ros-humble-tf2
ros-humble-tf2-bullet ros-humble-tf2-eigen ros-humble-tf2-eigen-kdl ros-humble-tf2-geometry-msgs
ros-humble-tf2-kdl ros-humble-tf2-msgs ros-humble-tf2-py ros-humble-tf2-ros ros-humble-tf2-ros-py
ros-humble-tf2-sensor-msgs ros-humble-tf2-tools ros-humble-tinyxml-vendor ros-humble-tinyxml2-vendor
ros-humble-tlsf ros-humble-tlsf-cpp ros-humble-topic-monitor ros-humble-tracetools ros-humble-trajectory-msgs
ros-humble-turtlesim ros-humble-uncrustify-vendor ros-humble-unique-identifier-msgs ros-humble-urdf
ros-humble-urdf-parser-plugin ros-humble-urdfdom ros-humble-urdfdom-headers ros-humble-visualization-msgs
ros-humble-yaml-cpp-vendor ros-humble-zstd-vendor rpcsvc-proto shiboken2 sip-dev tango-icon-theme tcl-dev
tcl8.6-dev tk tk-dev tk8.6 tk8.6-blt2.5 tk8.6-dev uncrustify unicode-data unixodbc-common unixodbc-dev uuid-dev
va-driver-all vdpau-driver-all vtk9 x11proto-dev xorg-sgml-doctools xtrans-dev zlib1g-dev
The following packages will be upgraded:
  cpp-11 gcc-11-base gcc-12-base libasound2 libasound2-data libatomic1 libblkid1 libc6 libc6-dbg libegl-mesa0
  libexif12 libexpat1 libfreetype6 libgbm1 libgcc-s1 libgif7 libgl1-mesa-dri libglapi-mesa libglv2.0-0
  libglv2.0-bin libglx-mesa0 libgomp1 libgphoto2-6 libgphoto2-l10n liblzm5 libmount1 libopenjp2-7 libpng16-16
  libpoppler-cpp0v5 libpoppler-glib8 libpoppler118 libpython3.10 libpython3.10-minimal libpython3.10-stdlib
  libsqlite3-0 libssl3 libstdc++6 libtiff5 libuuid1 libxml2 libxpm4 poppler-utils python3-pil python3-pkg-resources
  python3.10 python3.10-minimal
46 upgraded, 1027 newly installed, 0 to remove and 406 not upgraded.
Need to get 705 MB/720 MB of archives.
After this operation, 2,951 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Figure 22. ROS 2 Humble Desktop installation confirmation.

After the packages finished installing, the terminal returned to the command prompt without errors.

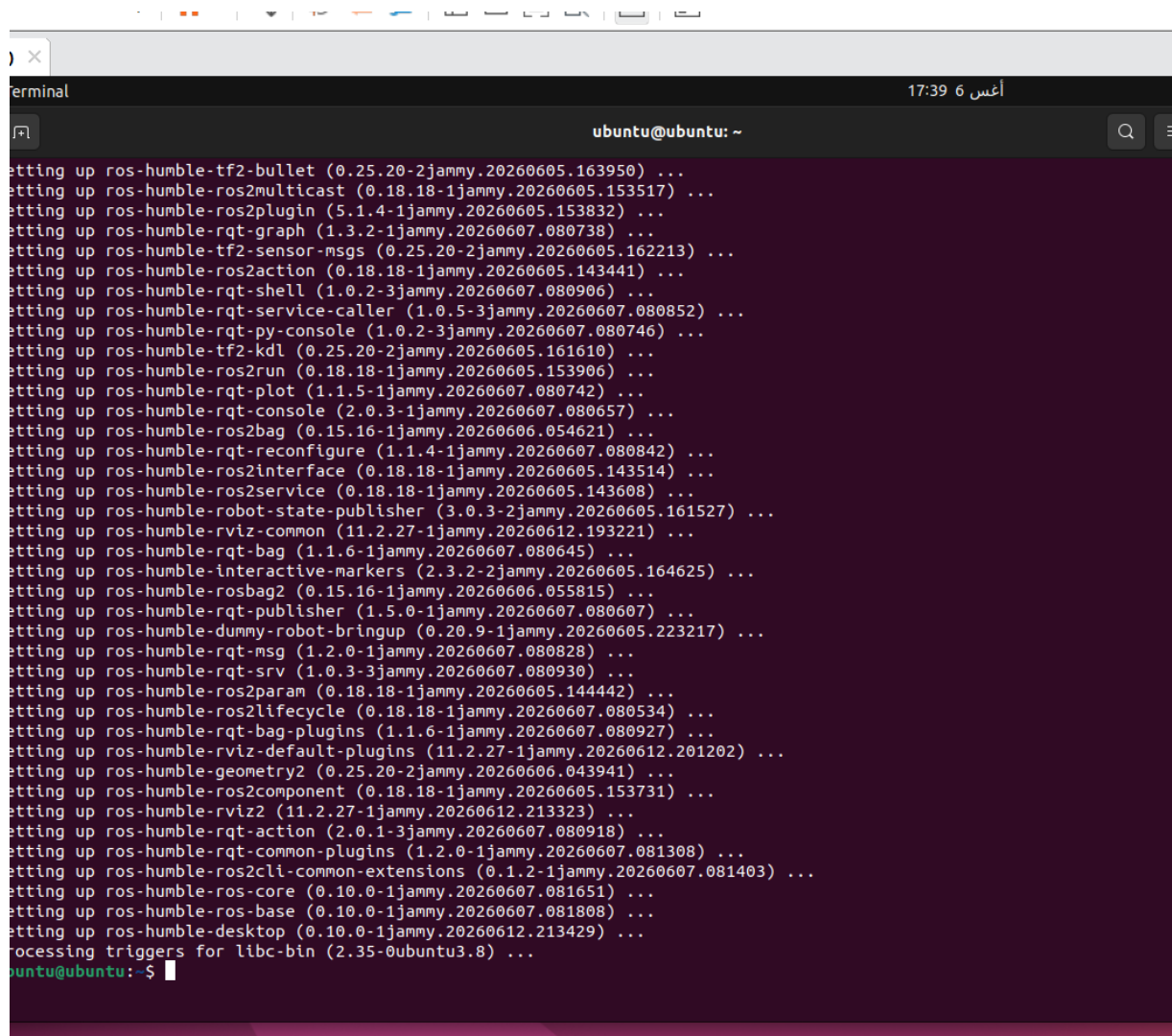
A terminal window titled 'terminal' with a dark background and light text. The window shows the output of the 'sudo apt-get install ros-humble-desktop' command. The output lists numerous ROS 2 packages being installed, each followed by its version and a series of dots indicating progress. The packages include ros-humble-tf2-bullet, ros-humble-ros2multicast, ros-humble-ros2plugin, ros-humble-rqt-graph, ros-humble-tf2-sensor-msgs, ros-humble-ros2action, ros-humble-rqt-shell, ros-humble-rqt-service-caller, ros-humble-rqt-py-console, ros-humble-tf2-kdl, ros-humble-ros2run, ros-humble-rqt-plot, ros-humble-rqt-console, ros-humble-ros2bag, ros-humble-rqt-reconfigure, ros-humble-ros2interface, ros-humble-ros2service, ros-humble-robot-state-publisher, ros-humble-rviz-common, ros-humble-rqt-bag, ros-humble-interactive-markers, ros-humble-roscpp, ros-humble-rqt-publisher, ros-humble-dummy-robot-bringup, ros-humble-rqt-msg, ros-humble-rqt-srv, ros-humble-ros2param, ros-humble-ros2lifecycle, ros-humble-rqt-bag-plugins, ros-humble-rviz-default-plugins, ros-humble-geometry2, ros-humble-ros2component, ros-humble-rviz2, ros-humble-rqt-action, ros-humble-rqt-common-plugins, ros-humble-ros2cli-common-extensions, ros-humble-ros-core, ros-humble-ros-base, ros-humble-desktop, and processing triggers for libc-bin. The terminal ends with the prompt 'ubuntu@ubuntu: ~\$'.

Figure 23. ROS 2 Humble Desktop installation completed.

9. Configuring and Running ROS 2 Humble

The ROS environment was loaded into the current terminal session with:

```
source /opt/ros/humble/setup.bash
```

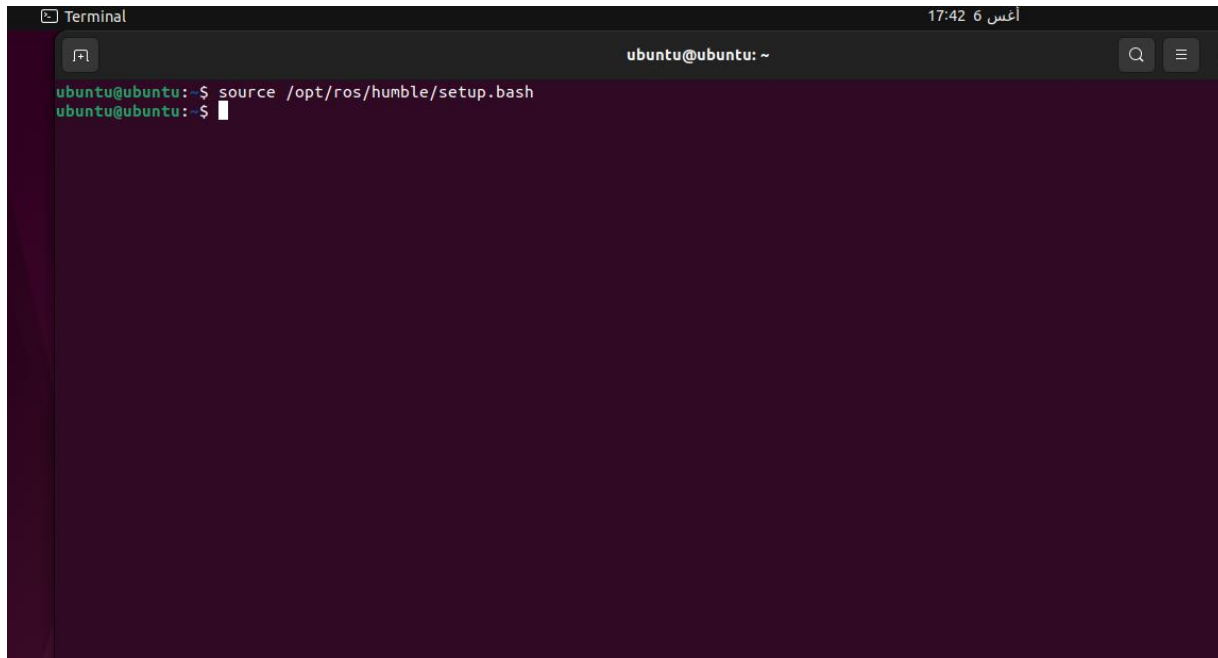


Figure 24. ROS 2 Humble environment sourced successfully.

To load ROS automatically whenever a new terminal is opened, the setup command was added to `.bashrc` and the file was reloaded:

```
echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc  
source ~/.bashrc
```

10. Final Verification

The ROS distribution environment variable was checked using:

```
echo $ROS_DISTRO
```

The terminal returned:

```
humble
```

This result confirms that ROS 2 Humble is installed, configured, and active.


```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ source /opt/ros/humble/setup.bash  
ubuntu@ubuntu:~$ echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc  
ubuntu@ubuntu:~$ source ~/.bashrc  
ubuntu@ubuntu:~$ echo $ROS_DISTRO  
humble  
ubuntu@ubuntu:~$
```

Figure 25. Final verification: ROS_DISTRO returns 'humble'.

11. Problems Encountered and Solutions

Problem	Cause	Solution
Username validation error during VMware Easy Install	The entered username contained unsupported formatting or characters.	A simple lowercase username was used: ubuntu.
Terminal prompt showed ubuntu@ubuntu-virtual-machine	VMware generated a long default hostname.	The hostname was changed using: sudo hostnamectl set-hostname ubuntu.
Mistyped add-apt-repository command	The command was entered incorrectly in the first attempts.	The correct command sudo add-apt-repository universe was entered and completed successfully.
software-properties-common did not install new files	The package was already installed.	No action was required because Ubuntu reported that the newest version was already installed.
ROS setup commands produced no visible output	source, echo-to-file, and some curl/tee commands can complete silently.	The next verification commands and the absence of errors were used to confirm success.
Large ROS installation size	ROS 2 Humble Desktop includes many tools and dependencies.	The required disk usage was reviewed before entering Y to continue.

12. Final Result

Ubuntu 22.04.5 LTS was successfully installed inside VMware Workstation. ROS 2 Humble Desktop was then installed from the ROS package repository and configured to load automatically. The final command 'echo \$ROS_DISTRO' returned 'humble', confirming that the task was completed successfully.

13. Commands Used

```
lsb_release -a
sudo hostnamectl set-hostname ubuntu
sudo apt update
locale
sudo apt install software-properties-common
sudo add-apt-repository universe
sudo apt install curl -y
sudo curl -sSL
https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o
/usr/share/keyrings/ros-archive-keyring.gpg
echo "deb [arch=$(dpkg --print-architecture) signed-
by=/usr/share/keyrings/ros-archive-keyring.gpg]
http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo
$UBUNTU_CODENAME) main" | sudo tee /etc/apt/sources.list.d/ros2.list >
/dev/null
sudo apt update
sudo apt install ros-humble-desktop
source /opt/ros/humble/setup.bash
echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc
source ~/.bashrc
echo $ROS_DISTRO
```

14. Conclusion

The task demonstrated the complete process of preparing a Linux environment for robotics development. Ubuntu 22.04.5 LTS was installed safely in a VMware virtual machine, the required Ubuntu repositories were prepared, ROS 2 Humble Desktop was installed, and the ROS environment was configured. The successful 'humble' output provides the final evidence that ROS 2 is ready for use.